

- ✓ Use Great Expectations (GE) for validation
- ✓ Use Dask for scalable cleaning

✓ Why Use Great Expectations *Before* Cleaning?

Great Expectations (GE) is a **data validation and profiling framework** built for this kind of scenario:

Benefit	What it Does
 Detect dirty data before cleaning	Find missing, malformed, or out-of-range
 Profile expected formats	Auto-infer expectations (e.g., float, range)
 Separate validation from cleaning	Makes cleaning logic more precise
 Generates Data Docs	Clean, shareable reports
 Integrates with Pandas, Dask, Spark	Works seamlessly in Python ecosystems

Recommended Pipeline

plaintext

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```
[ Raw CSV ]
  ↓
[Great Expectations]
(generate & validate expectations)
  ↓
[Dask Data Cleaning]
(use GE validation feedback)
  ↓
[Cleaned CSV / TFRecords / Parquet]
```

Summary

Task	Tool	
Validation/Profiling	 Great Expectations	
Scalable cleaning	 Dask	
Schema consistency checks	 GE	
Feature prep (TF-ready)	 TFT (after this)	